Comparative Analysis of Single-Activity Income Strategies in Stardew Valley

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***Abstract*—This study compares the economic performance of four exclusive income-generating activities (Farming, Fishing, Mining, and Foraging) in *Stardew Valley* across a full in-game year under time and energy constraints. Using 448 daily observations, Net Activity Income and Gold Per Minute (GPM) were analyzed with non-parametric tests due to non-normal distributions. Significant differences were found across activities for both income and efficiency ($p < 0.001$). Fishing generated the highest cumulative and mean daily income (923.72g) with low volatility, indicating strong stability. Farming achieved the highest median income (1,387.50g) and median GPM (8.46), reflecting superior time efficiency but greater variability. Mining showed moderate, episodic returns sensitive to luck conditions, while Foraging produced the lowest income and efficiency despite zero expenses. Results highlight trade-offs between stability, efficiency, and volatility in resource-constrained gameplay systems.**

***Index Terms*—simulation games, resource optimization, income efficiency, gameplay analytics, economic modeling, Stardew Valley.**

I. Introduction

Stardew Valley [1] is an open-ended, country-life role-playing game (RPG) that functions as a farming simulation game. At its core, the game presents a dilemma between traditional farm cultivation and the capitalistic expansion of the Joja Corporation. Beyond its "cozy" aesthetic, *Stardew Valley* operates as a simulation of real-world resource allocation and strategic planning. This study focuses on economic optimization and decision-making strategies within its virtual environment, specifically examining the financial efficiency of the four primary labor-based activities: farming, fishing, mining, and foraging. Governed by a rigid 20-hour daily cycle and a 112-day annual calendar, the game forces players to continually assess which activities yield the highest return on investment given limited time and energy resources. For players pursuing high-tier progression or in-game "perfection," identifying the most profitable strategy can determine whether progress stagnates or compounds exponentially. Analyzing these patterns provides insight into how individuals navigate multi-variable systems to achieve long-term objectives.

Prior investigations into game economies have largely relied on community-driven optimization guides and anecdotal evidence. While player-maintained resources such as wikis and forums offer theoretical yield estimates, academic literature on virtual economies suggests that such deterministic "solved game" approaches oversimplify player choice and strategic variability [2]. Moreover, models that overlook the stochastic nature of in-game systems fail to capture the real-time volatility introduced by factors such as daily luck and weather conditions [3]. To address these limitations, the present study employs a single-subject isolation design in which one full in-game year is dedicated exclusively to a single labor activity. By eliminating multitasking, this approach reduces observational noise and enables a clearer measurement of the empirical economic potential of each activity when pursued to its practical limit.

The primary objective of this study is to compare the distributional characteristics of daily net activity income and time-based efficiency (Gold Per Minute) across four single-activity gameplay strategies within a controlled one-year playthrough. Additionally, the study examines how environmental and progression-related variables influence financial outcomes. Through this empirical analysis, the research aims to provide data-driven insights that support strategic decision-making within time-constrained virtual economies.

II. Literature Review

This section reviews studies on economic optimization, resource allocation, and progression mechanics in simulation games, focusing on *Stardew Valley* and related frameworks in game design, virtual economies, and resource management.

A. Resource Allocation and Optimization

Stardew Valley presents a constrained virtual economy where players must manage finite time (6:00 AM–2:00 AM) and energy (270 units) [1]. De Weerd [4] applied Mixed Integer Linear Programming (MILP) to map actions, energy costs, and time blocks, demonstrating that optimal strategies could increase seasonal income by 146% over naive play. Automation tools, such as sprinklers, further enhance efficiency by freeing energy and time for higher-value activities, reflecting real-world capital investment theory where upfront costs yield long-term gains [4].

Game design and game theory research emphasize that constrained resources generate meaningful strategic choices. Sirlin [5] argues that opportunity costs are central to decision-making in competitive and resource-limited environments, while Salen and Zimmerman [6] describe how progression mechanics structure player engagement through trade-offs. In *Stardew*

Valley, each farming, fishing, or mining action represents a distinct cost in time and energy, forcing players to evaluate short-term returns versus long-term benefits. These principles demonstrate how simulation games function as controlled economic environments, where emergent behavior arises from the interplay between player decisions and systemic constraints.

B. Skill Progression and Economic Amplification

Skill progression systems in games create non-linear trajectories that amplify player capabilities and economic output over time. Adams and Dormans [7] distinguish between linear and exponential progression, noting that exponential or punctuated models produce dramatic strategic shifts. *Stardew Valley* exemplifies a hybrid system: experience points accumulate linearly, but major benefits occur at skill levels 5 and 10, such as reduced energy costs or profession bonuses. These discrete milestones produce strategic inflection points that fundamentally alter optimal approaches to farming, fishing, or mining.

Moreover, multiplicative bonuses and skill efficiencies create compounding returns that strongly influence long-term economic outcomes. Cook [8] explains how small incremental advantages in game systems can generate exponential growth in performance, a pattern observable in *Stardew Valley*'s late-game economy. For example, fishing energy reductions or the Artisan profession's 40% bonus to processed goods allow players to achieve higher yields with fewer resources, transforming early investment decisions into sustained advantages. Chen [9] notes that such progression systems balance challenge and player capability, maintaining engagement through an evolving risk-reward dynamic. These mechanisms show how skill systems in simulation games both guide player behavior and produce measurable effects on virtual economies.

C. Specialization, Diversification, and Virtual Economies

Economic theory provides insight into strategy selection between specialization and diversification in simulated environments. Ricardo's principle of comparative advantage [10] suggests that players should focus on activities where they have relative efficiency gains, while Varian [11] highlights the hidden costs of multitasking or activity switching. In *Stardew Valley*, fishing is low-cost and immediately profitable but cannot be automated, while farming requires upfront investment yet scales exponentially through automation and crop optimization [4]. These dynamics illustrate the trade-offs between immediate returns and long-term efficiency in virtual economies.

Diversification reduces risk but introduces management overhead, as predicted by portfolio theory [12]. Seasonal and environmental factors, such as Winter crop failure, rain-dependent fishing, and weather-independent mining, encourage multi-activity strategies, but managing multiple skills slows progression relative to specialization. De Weerd [4] quantifies this effect, showing that diversifying across activities incurs time penalties of 10–15% compared to focusing on a single skill. *Stardew Valley*'s deterministic systems and explicit rules create a tractable agent-based model of virtual economic behavior, making it a valuable testbed for studying comparative strategy, efficiency trade-offs, and the effects of discrete progression milestones [13].

D. Research Gap

Prior optimization studies primarily examine mixed-activity strategies within single seasons, leaving a gap in understanding the longitudinal effects of pure, single-activity approaches. This study addresses this gap by analyzing full-year gameplay, isolating individual economic mechanisms such as crop farming, fishing, or mining. Tracking skill and tool progression explicitly allows for a nuanced assessment of how discrete milestones affect economic efficiency, bridging the divide between theoretical optimization models and actual player performance.

By combining principles from game design, virtual economies, and resource management, this research provides a controlled framework to evaluate trade-offs between specialization, diversification, and skill development in *Stardew Valley*. The findings can inform both practical gameplay strategies and broader theoretical understanding of optimization in constrained, emergent economic systems.

III. Methodology

This section outlines the methodology used to collect, preprocess, and analyze data for the comparative evaluation of single income-generating activities in Stardew Valley. The study employed a quantitative approach to measure gold-per-minute efficiency and total net income under controlled gameplay conditions.

A. Participants

The study utilized a single-subject design where the researcher served as the sole participant. Demographic data were not collected as the research focus remained strictly on in-game strategic outcomes. Real-world dates of gameplay were recorded for documentation purposes but held no influence over the experimental variables.

B. Data Collection Methods

Data collection followed a dual-temporal approach, with real-time logging occurring weekly while in-game metrics were recorded daily. To capture the primary objective of the study, in-game variables were collected by manually logging in a physical diary during gameplay and subsequently encoded into Google Sheets for analysis. Each of the four primary activities was isolated and played for exactly one in-game year to ensure a controlled experimental environment.

The collected variables were categorized into four distinct domains, namely temporal, environmental, financial, and progressional. The temporal variables consist of the date, in-game date, and time duration. Environmental variables account for the seasonal changes, weather conditions, and daily luck. Financial metrics track gold in hand, primary activity gold, non-activity gold, and expenses. Lastly, the progressional

variables monitor the tool tiers and skill level advancements. These variables are essential in capturing player growth and gameplay performance.

C. Operational Definitions

- Date – The real-life calendar date on which the gameplay session was conducted.
- In-game Date – The corresponding in-game date, with each season consisting of 28 days and four seasons forming one in-game year, formatted as Day X, Year Y.
- Activity – The designated gameplay focus for the session, classified into four categories: farming, fishing, mining, and foraging.
- Season – The in-game seasonal period, divided into spring, summer, fall, and winter.
- Weather – The daily in-game atmospheric condition obtained from the television forecast, categorized into nine types: sunny, cloudy, windy (summer), windy (fall), rainy, thunderstorm, snow, green rain, and festival.
- Luck – The daily in-game statistic affecting drop rates and success probabilities, based on the description provided by the television fortune teller.
- Time Spent – The in-game duration spent performing the designated activity, recorded in hh:mm:ss format, with in-game time advancing in 10-minute increments.
- Gold on Hand – The total amount of gold currently possessed by the player.
- Primary Activity Gold – The total amount of gold earned from the designated activity.
- Non-Activity Gold – The total amount of gold earned from sources outside the designated activity, such as mail, quests, or geodes.
- Expenses – The total amount of gold spent by the player during the day.
- Tool Levels – The tier of tools acquired by the player, including the pickaxe, axe, sword, hoe, scythe, pan, fishing rod, and tackle, ranging from the base tier to iridium.
- Skill Levels – The level of each in-game skill, which increases as the player performs the corresponding activity, ranging from Level 1 to Level 10, with additional benefits granted at Levels 5 and 10.

D. Data Cleaning

To ensure accuracy and consistency, the dataset underwent systematic preprocessing prior to analysis. Time duration values were converted into total minutes to standardize the measurement of activity engagement. Numerical variables, including Gold on Hand, Primary Activity Gold, Non-Activity Gold, and Expenses, were converted to float data types to maintain computational precision and uniformity. Categorical variables such as Activity, Season, Weather, and Luck were cast to the "category" data type to preserve their discrete structure and support proper visualization and analysis. For clearer interpretation of environmental effects, the "Cloudy" weather category was reclassified into Windy (Summer) and Windy (Fall) according to seasonal context. Ordinal relationships were explicitly defined for Season and Luck to maintain logical ordering. Similarly, tool upgrades were encoded using ordered categories to reflect progression tiers. The pickaxe, axe, hoe, and scythe were categorized as Base, Copper, Iron, Gold, and Iridium, while the fishing rod was ordered from Training, Bamboo, Fiberglass, Iridium, to Advanced Iridium.

Additional feature engineering was performed to enhance analytical depth. The In-game Date variable was decomposed to extract Game Year and Game Day attributes, and a Continuous Game Day variable was created to preserve chronological sequencing across observations. Two key financial indicators were derived to align with the study objectives. Net Activity Income was calculated as primary activity gold minus daily expenses, representing actual daily profit. Gold Per Minute (GPM) was computed by dividing net activity income by total in-game minutes, serving as the primary measure of time-based efficiency. These derived metrics formed the core variables used in comparative and statistical analyses. Finally, non-contributory columns, including Date, In-game Date (after feature extraction), and Pan, were removed after confirming that they did not influence financial or efficiency outcomes.

E. Statistical Analysis

Statistical analysis was conducted to evaluate differences in Net Activity Income and Gold Per Minute (GPM) across the four income-generating activities: Farming, Fishing, Mining, and Foraging. The distribution of Net Activity Income and Gold Per Minute within each activity group was assessed using the Shapiro–Wilk normality test. The test statistic is defined as

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \tag{1}$$

where $x_{(i)}$ denotes the ordered sample values, $\bar{x}$ represents the sample mean, a_i are constants derived from the expected order statistics of a normal distribution, and n is the sample size. For each activity group, the null hypothesis tested was H_0: the data are normally distributed. Statistical significance was evaluated at the $\alpha = 0.05$ level.

To compare income performance across activities, the Kruskal–Wallis H test was employed. The hypotheses for Net Activity Income were defined as

- H_{0_1}: There is no difference in net activity income among the income-generating activities.
- H_{1_1}: At least one income-generating activity has a net activity income different from the others.

Similarly, the hypotheses for Gold Per Minute were defined as

- H_{0_2}: There is no difference in gold-per-minute (GPM) efficiency among the income-generating activities.
- H_{1_2}: At least one income-generating activity differs in gold-per-minute (GPM) efficiency compared to the others.

The Kruskal–Wallis test statistic is computed as

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} n_i \bar{R}_i^2 - 3(N+1) \quad (2)$$

where N is the total number of observations, k is the number of activity groups, n_i is the sample size of group i, and $\bar{R}_i$ is the mean rank of group i. Statistical significance was evaluated at the $\alpha = 0.05$ level.

When the Kruskal–Wallis test indicated statistical significance, Dunn's post-hoc test was performed to identify pairwise differences between activity groups. Dunn's test compares mean rank differences between groups, and the Bonferroni correction was applied to control for multiple comparisons according to

$$\alpha_{\text{adjusted}} = \frac{\alpha}{m} \quad (3)$$

where m represents the number of pairwise comparisons. Adjusted p-values were used to determine statistical significance for each activity pair. All statistical analyses were performed using Python statistical libraries, and test statistics together with corresponding p-values were reported.

The results were presented through several visualizations created using the Matplotlib and Seaborn libraries. These libraries were utilized to generate boxplots, line charts, distribution plots, and correlation matrices to illustrate income trends, efficiency metrics, and relationships between gameplay variables. Boxplots were used to examine the distribution and variability of financial metrics across activities. Line charts were employed to visualize temporal trends, including daily and cumulative income performance over continuous game days. Distribution plots were used to compare overall performance patterns across activities, while heatmaps were generated to present correlation relationships between categorical environmental variables and financial outcomes. These visualization techniques enabled clear comparative analysis and facilitated the identification of performance patterns, variability, and trends within the dataset.

IV. Results

This section presents the quantitative findings from the comparative analysis of single income-generating activities in Stardew Valley. The objective of this analysis is to determine which of the labor-based activities generated the highest daily net activity income and is the most efficient in terms of gold-per-minute when played exclusively in one year.

Following data preprocessing, the final dataset contained 448 observations across 26 variables. Several derived features were introduced to align with the study objectives, including Game Year and Day, Continuous Game Day, Net Activity Income, and Gold Per Minute. These engineered variables enabled longitudinal performance tracking and standardized efficiency comparisons across activities.

The analysis primarily compared income-generating activities using net activity income and gold-per-minute (GPM) as the key performance metrics. Additionally, the study examined the relationships between categorical gameplay variables, specifically weather, season, luck, and tool upgrades, and their influence on financial performance. Skill progression levels were also analyzed to evaluate their impact on income and efficiency outcomes, providing further context for activity-based performance differences.

TABLE I
Summary of Dataset Columns

Column	Count	Dtype
Activity	448	Category
Season	448	Category
Weather	448	Category
Luck	448	Category
Time Spent (in minutes)	448	Int
Gold on Hand	448	Float
Primary Activity Gold	448	Float
Non-Activity Gold	448	Float
Expenses	448	Float
Pickaxe	448	Category
Axe	448	Category
Sword	224	Category
Scythe	448	Category
Rod	331	Category
Tackle	7	Category
Farming Lvl.	448	Int
Fishing Lvl.	448	Int
Mining Lvl.	448	Int
Foraging Lvl.	448	Int
Fighting Lvl.	448	Int
Game Year	448	Int
Game Day	448	Int
Continuous Game Day	448	Int
Net Activity Income	448	Float
Gold Per Minute	448	Float

TABLE II
Net Activity Income Summary Across Activities

Activity	Mean	Median	SD
Farming	840.25	1387.5	3250.40
Fishing	923.72	1311.5	2013.91
Foraging	260.38	182.5	258.77
Mining	817.12	525.0	2025.75

Table II presents the summary statistics of Net Activity Income across the four activities. Fishing recorded the highest mean Net Activity Income at 923.72g, while Foraging recorded the lowest at 260.38g. Farming had the highest variability in income, with a standard deviation of 3250.40g. Mining showed a lower median income (525.0g) relative to its mean (817.12g), suggesting skewed income distribution in mining days.

TABLE III
PRIMARY ACTIVITY GOLD SUMMARY ACROSS ACTIVITIES

Activity	Mean	Median	SD
Farming	1839.27	1755.0	1828.98
Fishing	1271.49	1358.5	760.72
Foraging	260.38	182.5	258.77
Mining	1432.74	999.0	1535.81

Table III summarizes Primary Activity Gold across the four activities. Farming achieved the highest mean Primary Activity Gold at 1839.27g, followed by Mining at 1432.74g. Foraging again recorded the lowest values (260.38g). Fishing had a mean of 1271.49g and the smallest standard deviation (760.72g), indicating relatively consistent gold earnings from fishing days.

TABLE IV
EXPENSES, GOLD PER MINUTE (GPM), AND TIME SPENT SUMMARY ACROSS ACTIVITIES

Activity	Expenses Mean	GPM Mean	Time Spent Mean (min)
Farming	999.02	5.43	142.50
Fishing	347.77	2.69	338.12
Foraging	0.00	0.63	394.38
Mining	615.62	2.46	226.61

Table IV presents the summary of Expenses, Gold Per Minute (GPM), and Time Spent across activities. Farming recorded the highest mean Expenses at 999.02g and achieved the highest mean GPM at 5.43. Fishing followed with a GPM of 2.69, Mining 2.46, and Foraging the lowest at 0.63. Foraging recorded zero expenses across all observations and had the longest average time spent per day at 394.38 minutes, while Farming had the shortest at 142.50 minutes.

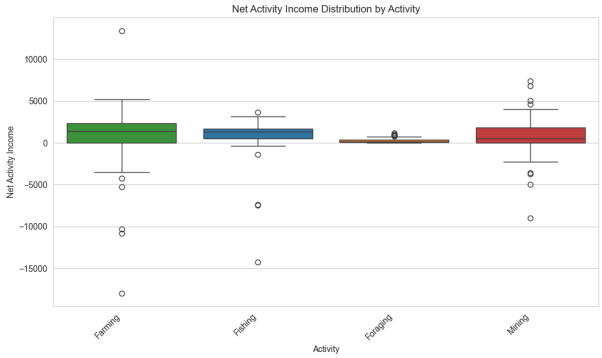

Fig. 1. Net Activity Income Distribution by Activity

Fig. 1 showed the distribution of daily Net Activity Income across activities. Farming exhibited the widest range, with outliers exceeding 14,000g and losses reaching approximately –18,000g. Foraging showed the narrowest distribution, with most values concentrated between 180g and 260g.

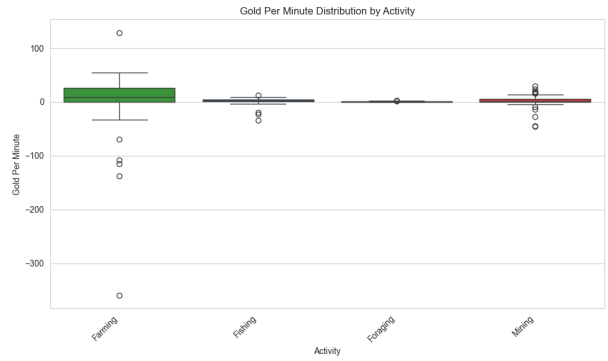

Fig. 2. Gold Per Minute Distribution by Activity

Fig. 2 illustrated the distribution of Gold Per Minute across activities. Farming recorded the highest median GPM at approximately 5.43, with the widest variability range. Foraging recorded the lowest GPM values, centered around 0.63, with minimal dispersion.

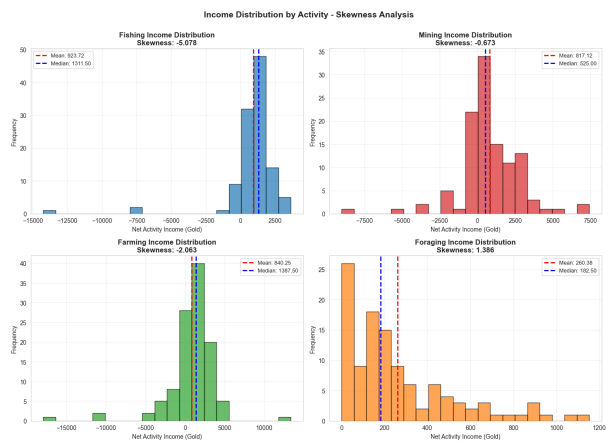

Fig. 3. Skewness Analysis: Net Activity Income by Activity

Fig. 3 presents the income distributions for Fishing, Mining, Farming, and Foraging, including their corresponding skewness values, means, and medians. Fishing (skewness $= -5.078$), Farming (2.063), and Mining (0.673) exhibit negative skewness, with visible lower-end outliers and medians generally higher than means. In contrast, Foraging (1.386) shows positive skewness, with a longer right tail and a mean exceeding the median. The histograms illustrate varying spreads across activities, with Farming and Fishing displaying wider ranges of net income compared to Mining and Foraging.

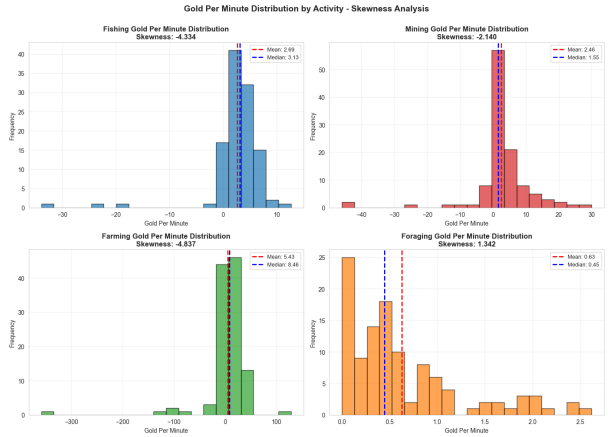

Fig. 4. Skewness Analysis: Gold Per Minute by Activity

Fig. 4 shows variations in gold-per-minute (GPM) values across the four activities. Farming and mining exhibit wider ranges of GPM values compared to fishing and foraging. Fishing and farming display higher median values than their corresponding means, while mining and foraging show higher mean values than their medians. Among the four activities, farming demonstrates the largest spread of values, while foraging shows the narrowest concentration of GPM values.

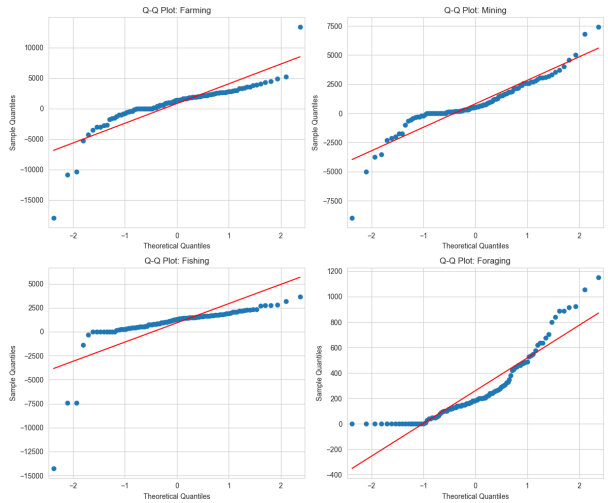

Fig. 5. Q-Q Plot: Net Activity Income by Activity

Fig. 5 shows that the data points deviate substantially from the theoretical normal line, particularly in the lower and upper tails. This indicates that the distribution of Net Activity Income is skewed and not normally distributed across activities.

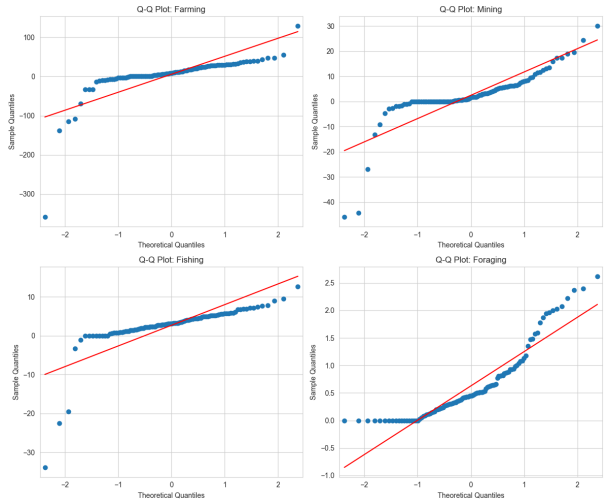

Fig. 6. Q-Q Plot: Gold Per Minute by Activity

Fig. 6 reveals noticeable departures from the diagonal line, with pronounced deviations at both tails. This pattern indicates non-normality in the distribution of GPM values across activities

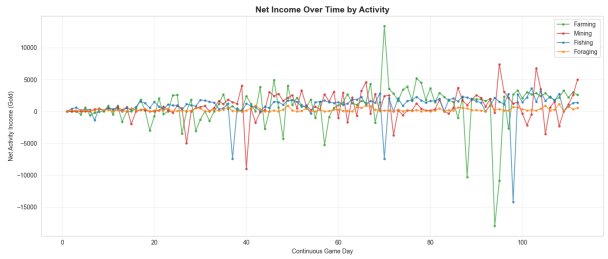

Fig. 7. Net Activity Income Over Time by Activity

Fig. 7 presented daily Net Activity Income across 112 continuous game days per activity. Farming displayed noticeable fluctuations, alternating between negative and positive income values. Foraging maintained relatively stable daily income levels averaging below 300g.

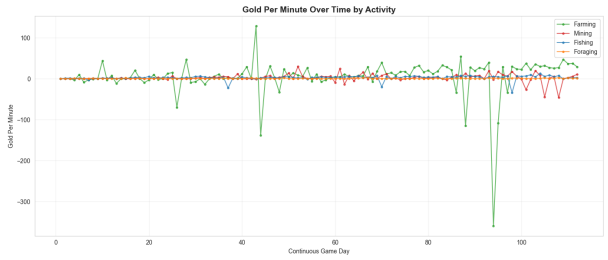

Fig. 8. Gold Per Minute Over Time by Activity

Fig. 8 illustrated daily GPM trends across continuous gameplay days. Farming showed peaks exceeding 5 GPM during harvest periods and declines during planting phases. Fishing, Mining, and Foraging maintained relatively consistent GPM values, generally remaining below 3 GPM.

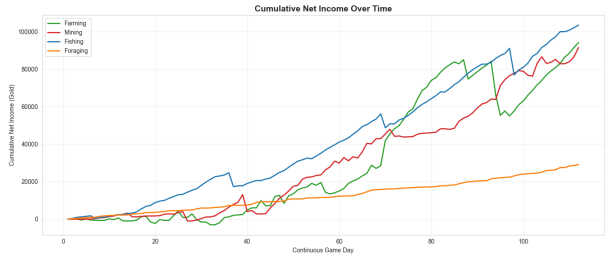

Fig. 9. Cumulative Net Activity Over Time

Fig. 9 showed cumulative Net Activity Income across the in-game year. Fishing demonstrated steady income accumulation, exceeding 100,000g by the end of the observation period. Farming showed slower early accumulation followed by gradual increases across later game days.

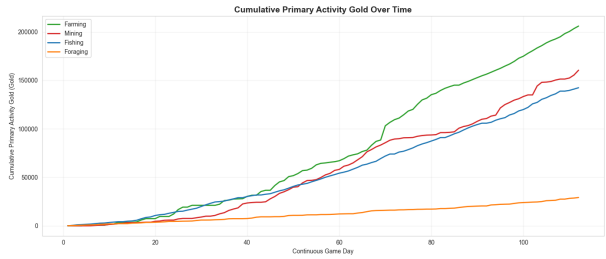

Fig. 10. Cumulative Primary Activity Gold Over Time

Fig. 10 compared cumulative Primary Activity Gold across activities. Farming recorded the highest cumulative gross earnings, exceeding 180,000g. Fishing and Mining recorded lower cumulative totals, while Foraging remained below 30,000g.

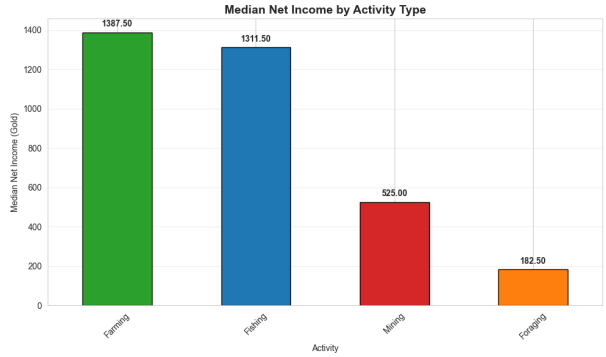

Fig. 11. Average Net Activity Income by Activity Type

Fig. 11 illustrates the distribution of median Net Activity Income among activities. Farming recorded the highest median net income at 1,387.50g, closely followed by Fishing at 1,311.50g. Mining produced a substantially lower median income of 525.00g, while Foraging remained the least profitable activity with 182.50g.

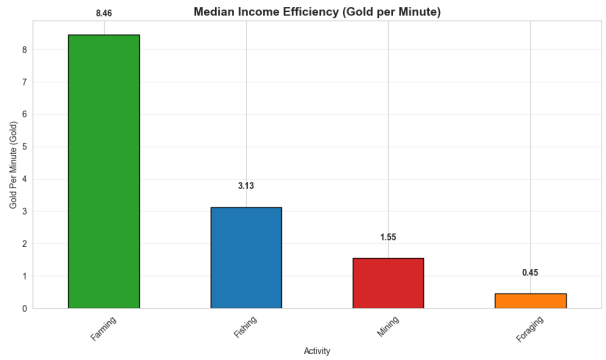

Fig. 12. Average Income Efficiency (Gold Per Minute)

Fig. 12 compares the median Gold Per Minute (GPM) across activities. Farming demonstrated the highest efficiency at 8.46 GPM, significantly outperforming other activities. Fishing followed at 3.13 GPM, while Mining achieved 1.55 GPM. Foraging recorded the lowest efficiency at 0.45 GPM.

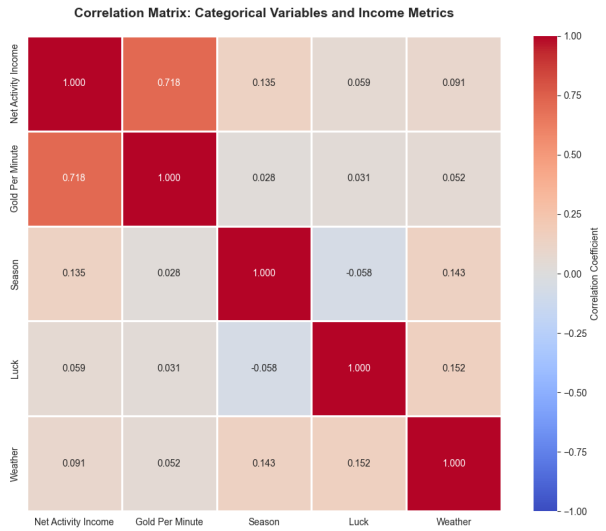

Fig. 13. Correlation Matrix: Categorical Variables and Income Metrics

Fig. 13 presented the correlation matrix between weather, season, luck, Net Activity Income, and Gold Per Minute. The matrix showed weak positive correlations between environmental variables and the financial performance metrics.

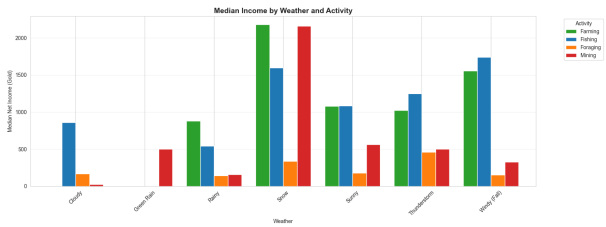

Fig. 14. Median Net Activity Income by Weather and Activity

Fig. 14 presents median Net Activity Income across different weather conditions. Snow generated the highest incomes

for both Farming (2,180g) and Mining (2,160g), with Fishing (1,590g) also performing strongly. Windy (Fall) conditions favored Fishing (1,730g) and Farming (1,550g). Thunderstorms particularly benefited Fishing (1,240g).

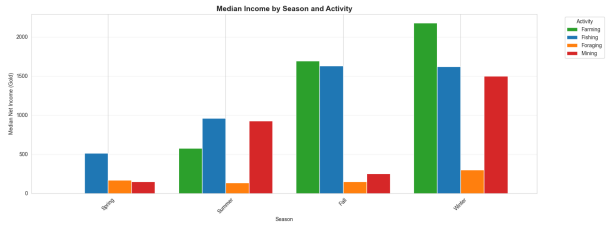

Fig. 15. Median Net Activity Income by Season and Activity

Fig. 15 shows seasonal median income comparisons across activities. Fishing produced the highest median income in Spring and Summer, while Farming produced the highest median income in Fall and Winter. Mining showed strong seasonal swings, with its highest median income in Winter and a secondary peak in Summer, but much lower values in Spring and Fall. Foraging remained the lowest-earning activity across all seasons, staying below roughly 300g in every season.

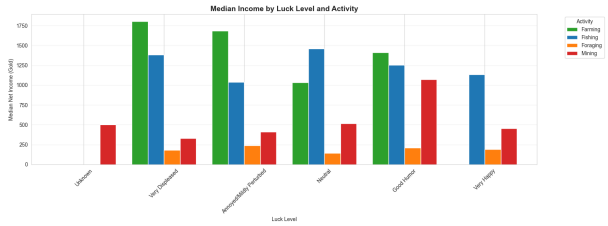

Fig. 16. Median Net Activity Income by Luck and Activity

Fig. 16 illustrates median net activity income across luck levels. Farming and Fishing remain relatively stable across luck conditions, with only modest variation. Mining, however, shows the strongest sensitivity to luck, reaching its highest median income during the "Good Humor" condition (around 1,050g) and lower median outcomes under less favorable luck states. Foraging remains consistently low across all luck levels.

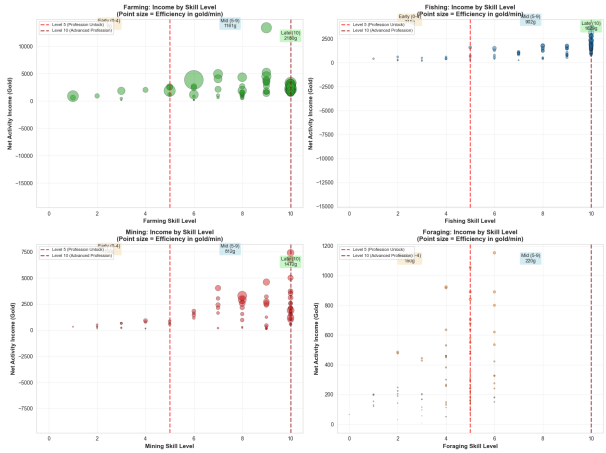

Fig. 17. Net Activity Income over Activity Skill Levels

Fig. 17 shows net activity income by skill level (0–10) for Farming, Fishing, Mining, and Foraging, with bubble size representing gold-per-minute efficiency. Across all activities, income generally increases with skill level, and larger bubbles are concentrated in the mid-to-late range (Levels 5–10), indicating improved efficiency after profession milestones. Farming and Mining exhibit the widest spread of outcomes at higher levels, including occasional very high returns, while Fishing shows a tighter upward trend. Foraging remains comparatively low across all levels, with only moderate gains even at higher skill levels.

TABLE V
SHAPIRO-WILK NORMALITY TEST

Activity	Net Activity Income		Gold Per Minute	
	W-statistic	P-value	W-statistic	P-value
Farming	0.7637	3.87×10^{-12}	0.5551	7.55×10^{-17}
Fishing	0.5121	1.32×10^{-17}	0.5761	1.85×10^{-16}
Mining	0.8901	1.41×10^{-7}	0.7197	2.51×10^{-13}
Foraging	0.8499	2.76×10^{-9}	0.8468	2.10×10^{-9}

Table V presents the results of the Shapiro-Wilk normality test for Net Activity Income and Gold Per Minute across different activities. For Net Activity Income, the W-statistics ranged from 0.5121 (Fishing) to 0.8901 (Mining), with all corresponding p-values below 0.05, indicating that the assumption of normality was violated for all activities. Similarly, for Gold Per Minute, the W-statistics ranged from 0.5551 (Farming) to 0.8468 (Foraging), and all p-values were also below 0.05, suggesting that the distributions for Gold Per Minute across activities deviate significantly from normality.

TABLE VI
KRUSKAL-WALLIS TEST

Measure	H-statistics	P-value
Net Activity Income	52.91	1.91×10^{-11}
Gold Per Minute	57.32	2.19×10^{-12}

Table VI presents the results of the Kruskal-Wallis H-test for Net Activity Income and Gold Per Minute across different activities. The analysis revealed a statistically significant difference in Net Activity Income between activities ($H = 52.91, p < 0.001$), indicating that at least one activity differs in earnings compared to the others. Similarly, Gold Per Minute also showed a significant difference across activities ($H = 57.32, p < 0.001$), suggesting variation in time-based efficiency between activities. These results support the presence of meaningful differences in both income and efficiency measures across the activity groups.

TABLE VII
DUNN POST-HOC TEST: NET ACTIVITY INCOME

Activity	Farming	Fishing	Foraging	Mining
Farming	1.00×10^{0}	1.00×10^{0}	1.40×10^{-7}	0.7049
Fishing	1.00×10^{0}	1.00×10^{0}	5.23×10^{-11}	0.0301
Foraging	1.40×10^{-7}	5.23×10^{-11}	1.00×10^{0}	0.0003
Mining	7.05×10^{-1}	3.01×10^{-2}	3.49×10^{-4}	1.0000

Table VII presents Dunn post-hoc pairwise comparisons for Net Activity Income. Significant differences were observed between Foraging and all other activities, as well as between Mining and Fishing, indicating that specific activity pairs differ meaningfully in earnings.

TABLE VIII
DUNN POST-HOC TEST: GOLD PER MINUTE

Activity	Farming	Fishing	Foraging	Mining
Farming	1.00×10^{0}	1.00×10^{0}	6.09×10^{-11}	0.0022
Fishing	1.00×10^{0}	1.00×10^{0}	8.31×10^{-9}	0.0296
Foraging	6.09×10^{-11}	8.31×10^{-9}	1.00×10^{0}	0.0070
Mining	2.24×10^{-3}	2.96×10^{-2}	7.03×10^{-3}	1.0000

Table VIII presents Dunn post-hoc pairwise comparisons for Gold Per Minute. Foraging differed significantly from Farming and Fishing, while Mining showed significant differences from Farming, Fishing, and Foraging, highlighting meaningful variation in time-based efficiency across activities.

V. DISCUSSION

This section interprets the results of the study, examining how different in-game activities influence daily income and efficiency. Findings are discussed in relation to previous research on gameplay strategies and virtual economies, highlighting patterns, trade-offs, and player decision-making behavior. This analysis provides insight into optimizing performance within time-constrained game scenarios.

A. Interpretation of Results

The findings of this study provide comparative insights into the daily economic performance of single-activity strategies within a one-year gameplay period in *Stardew Valley*. Each activity generated 112 daily observations, and statistical comparisons were conducted on the distributions of daily Net Activity Income and Gold Per Minute (GPM) across activity groups.

The analysis indicates that the four activities exhibit distinct distributional characteristics in terms of profitability, stability, scalability, and time efficiency. Farming recorded the highest median Net Activity Income at 1,387.50g and the highest median GPM at 8.46, indicating superior daily performance during productive periods. However, its mean Net Activity Income (840.25g) fell below its median, reflecting the substantial downward pressure of negative-income days, particularly those occurring outside harvest windows or during Snow weather. The wide standard deviation of 3,250.40g and the presence of outliers exceeding 14,000g as well as losses reaching approximately $-18{,}000$g underscore Farming's high day-to-day volatility. Despite this variability, Farming accumulated the highest cumulative Primary Activity Gold over the observed year, exceeding 180,000g, indicating that its gross earnings potential is the strongest of all activities when favorable conditions align. These patterns position Farming as a high-efficiency but high-volatility strategy whose daily performance is strongly shaped by the cyclical structure of crop investment and harvest returns.

Fishing produced the highest mean Net Activity Income at 923.72g and the highest cumulative Net Activity Income across the observation period, exceeding 100,000g by the end of the in-game year. Its comparatively low expense mean (347.77g) and relatively small standard deviation in Primary Activity Gold (760.72g) indicate that Fishing provides a stable and consistent income stream under exclusive-play conditions. Unlike Farming, Fishing does not depend on seasonal planting windows or harvest cycles, which contributes to steadier daily accumulation. These characteristics suggest that Fishing functions as a low-capital, low-volatility strategy, making it well-suited for players prioritizing income stability over peak-day efficiency.

Mining displayed moderate daily performance across financial metrics. Its mean Net Activity Income (817.12g) was close to Farming's, yet its median (525.00g) was substantially lower, reflecting a positively skewed income distribution driven by episodic gains from resource variability and progression milestones. Seasonal patterns indicate relatively stronger Mining performance during Summer and Winter, though these differences represent contextual daily tendencies rather than replicated seasonal experiments. Mining also showed the strongest sensitivity to luck conditions among all activities, reaching its highest median income (approximately 1,050g) under "Good Humor" luck, consistent with the stochastic reward structures that characterize extractive gameplay activities.

Foraging consistently recorded the lowest daily Net Activity Income (mean: 260.38g; median: 182.50g) and the lowest mean GPM (0.63) across all activities. Although it incurred zero expenses and demonstrated relatively stable daily outcomes, its income distribution remained concentrated within a narrow range, indicating limited scalability as a primary income strategy under exclusive conditions. Foraging also demanded the highest average daily time commitment at

394.38 minutes, making it not only the least profitable but also the most time-intensive activity relative to its returns. These findings suggest that while Foraging may complement other activities, its value as a standalone strategy is limited within a constrained gameplay environment.

The statistical analyses further support the presence of distributional differences across activities at the daily level. The Shapiro–Wilk tests confirmed non-normality in both Net Activity Income and GPM distributions for all four activities, justifying the use of non-parametric methods. The Kruskal–Wallis tests detected significant differences in daily income ($H = 52.91$, $p < 0.001$) and efficiency distributions ($H = 57.32$, $p < 0.001$) across activities. Dunn post-hoc comparisons further identified that Foraging differed significantly from all other activities in both metrics, while Mining differed significantly from Fishing in Net Activity Income and from Farming, Fishing, and Foraging in GPM. These results indicate that the observed variations in daily profitability and time efficiency are unlikely to be attributable solely to random fluctuation within this experimental run.

However, because each activity was performed in a single full-year sequence by the same player, daily observations may exhibit temporal dependence due to skill progression, tool upgrades, and cumulative economic effects. Therefore, the statistical comparisons should be interpreted as analyses of daily income distributions within this controlled gameplay context rather than as population-level generalizations across independent yearly replications.

Skill progression analysis revealed a consistent upward trend in daily income and efficiency as skill levels increased across all activities. Higher-income observations clustered more frequently around Levels 5 and 10, corresponding to profession milestone unlocks in the game's progression system. Farming and Mining exhibited the widest spread of outcomes at higher levels, including occasional very high returns, while Fishing showed a tighter upward trend. Foraging remained comparatively low across all levels, with only moderate income gains even at the maximum skill level. These patterns suggest that progression mechanics are associated with improved daily economic performance over time, reflecting compounding efficiency effects within the structured gameplay system.

Environmental variables such as season, weather, and luck demonstrated generally weak correlations with daily financial metrics, although activity-specific sensitivities were observed. Snow weather generated the highest median Net Activity Income for Farming (approximately 2,180g) and Mining (approximately 2,160g), while Windy (Fall) conditions favored Fishing (approximately 1,730g). Thunderstorm conditions particularly benefited Fishing due to in-game mechanics that improve catch quality during such weather. These findings suggest that contextual modifiers influence daily income variability, but their overall contribution appears secondary to structural differences between activity types within this experimental framework.

Overall, the results highlight meaningful trade-offs between cumulative stability, daily volatility, and time-based efficiency across the four activities. Farming maximizes time-efficiency and median daily income on productive days but carries significant downside risk. Fishing provides the most stable cumulative returns with the lowest expense burden. Mining offers moderate income with episodic gains, while Foraging presents the lowest profitability and efficiency under exclusive-play conditions. Rather than identifying a universally dominant strategy, the findings emphasize that optimal activity selection depends on the player's tolerance for income volatility and their prioritization of stability versus peak-day performance.

B. Comparison to Related Work

The results of this study are consistent with existing optimization research emphasizing resource constraints as central determinants of player strategy. Prior work using Mixed Integer Linear Programming models demonstrated that optimized gameplay strategies outperform naïve approaches by efficiently allocating time and energy resources [4]. The activity-specific differences observed in the distributions of daily Net Activity Income and GPM reflect how different income-generating activities exhibit distinct performance profiles under identical resource constraints, an outcome consistent with the broader principle that constrained resource systems generate strategic differentiation [6].

Farming's combination of the highest median Net Activity Income (1,387.50g) and the highest median GPM (8.46) reflects superior time-based productivity on harvest days. These metrics, however, coexist with substantial daily volatility, a pattern consistent with Markowitz's portfolio theory, wherein higher expected returns are typically accompanied by higher variance [12]. From a game design perspective, this volatility is a structural feature of the crop cycle mechanic rather than an artifact of player behavior, aligning with frameworks that describe how rule-based game systems generate emergent risk-reward trade-offs [6], [7]. The negative skewness observed in Farming's Net Activity Income distribution (-2.063) further illustrates how lower-tail outcomes incurred during non-harvest days and adverse weather significantly temper the activity's mean performance relative to its median.

Fishing's position as the highest mean-income and highest cumulative-income activity aligns with economic principles of comparative advantage [10]. Its lower expense profile (mean: 347.77g versus Farming's 999.02g), consistent daily returns, and strong negative skewness (-5.078), indicating a distribution concentrated in the upper income range with limited downside outliers, position Fishing as a low-capital, high-consistency strategy. This is consistent with theoretical perspectives on risk-averse resource allocation in stochastic game environments, where strategies yielding stable returns are preferable under income uncertainty [3], [11]. The contrast between Fishing's cumulative superiority and Farming's higher median daily performance illustrates how aggregate outcomes in a virtual economy can diverge from single-day efficiency metrics over an extended observation period.

Mining's income distribution exhibited mild negative skewness (-0.673), with its mean (817.12g) exceeding its median (525.00g), indicating that episodic high-yield days drive aggregate performance. This is consistent with stochastic reward structures observed in extractive activities, where resource variability introduces significant day-to-day unpredictability [3]. Mining's heightened sensitivity to luck conditions further reflects the role of probabilistic game mechanics in shaping extractive activity outcomes, a dynamic that prior work on in-game optimization has attributed to random loot and drop-rate systems [4]. Its moderate but significant GPM difference from Farming and Fishing, confirmed by Dunn post-hoc comparisons, positions Mining as a middle-ground activity with moderate efficiency and episodic income potential.

The consistent underperformance of Foraging across all metrics, including income, efficiency, and scalability, illustrates the trade-offs inherent in zero-cost strategies within a constrained gameplay economy. While Foraging avoids expenditure entirely, its narrow income range, low GPM (mean: 0.63; median: 0.45), and high time demand (mean: 394.38 minutes per day) reflect an opportunity cost that outweighs its expense-free appeal [11]. This outcome is consistent with the economic principle that resource efficiency, not merely cost avoidance, determines strategic value within constrained systems [5]. The positive skewness of Foraging's income distribution (1.386) suggests that while typical daily returns are modest, occasional higher-yield days contribute disproportionately to its aggregate total, a characteristic common to opportunistic resource-gathering strategies.

The positive association between skill progression and daily income across all activities aligns with progression system theories, in which milestone-based bonuses generate discrete improvements in player capability [7]. The clustering of higher-income and higher-GPM observations around skill Levels 5 and 10, corresponding to profession selection milestones in *Stardew Valley* [1], suggests that the game's progression architecture is deliberately designed to create measurable inflection points in economic performance. This mirrors feedback loop dynamics described in game chemistry literature, wherein incremental player investment compounds into qualitatively distinct capability states over time [8], [9].

The single-activity isolation design of this study provides a controlled context for examining daily performance differences across specialized strategies, extending prior optimization research [4] by documenting how daily economic behavior evolves alongside progression, tool upgrades, and contextual modifiers within a structured gameplay environment [13]. The theorcrafting literature further frames this type of controlled comparative analysis as a means of surfacing the strategic efficiency trade-offs embedded in game rule systems [2], and the present findings are consistent with that perspective: no single activity dominates across all metrics simultaneously.

Overall, these findings support theoretical frameworks in both game design and economic modeling, suggesting that different activities exhibit distinct daily performance profiles shaped by volatility, expense structure, time efficiency, and contextual modifiers. The results highlight the importance of aligning player objectives with activity-specific risk-reward profiles, reinforcing the view that optimal strategy selection in constrained virtual economies depends as much on income stability preferences as on peak-day efficiency potential [3], [5], [12].

C. Limitations

The limited duration of data collection inherently constrained the experiment's ability to fully capture the potential of each income-generating activity in Stardew Valley. Several advanced strategies and seasonal opportunities are only unlocked after completing at least one full in-game year, making it impossible to access and evaluate these within the one-year timeframe. Similarly, certain in-game bundles, which unlock supplementary tools, crops, or animals that could enhance income generation, could not be completed within the study period.

Tool upgrades and skill progression were not fully optimized because the experiment primarily focused on maximizing short-term income through a single activity rather than balanced long-term growth. Additionally, the experiment was restricted to the main island, limiting exposure to other islands that offer unique opportunities and advantages for certain activities such as exclusive crops, resources, or fishing locations. The variability of strategies was also constrained because activities were executed only once per season, which may not reflect typical player behavior or seasonal optimization.

Furthermore, the manual data collection method introduces potential inconsistencies that could impact the accuracy of the findings. Minor variations in gameplay such as differences in time spent, stamina consumption, or incidental events were not fully captured, leading to potential errors in the dataset. This method also risks introducing self-report bias, where subjective judgments or human error may skew the results.

These limitations suggest that the results may not fully reflect the maximum income potential or the optimal strategies for each activity. Consequently, conclusions about the comparative efficiency of the different activities should be interpreted cautiously, as they are based on partial progression, limited seasonal repetition, and isolated island conditions.

D. Recommendations and Future Work

To further explore the virtual economy in Stardew Valley, future research should incorporate additional gameplay variables to better capture the complexity of in-game decision-making. Stamina, a critical constraint on player activity, should be tracked because it directly affects the efficiency and feasibility of each income-generating strategy, particularly in the early days of the game. Recording the specific items responsible for generating income in each activity would allow a more detailed analysis of financial performance and resource optimization.

Extending the experiment beyond a single in-game year is recommended to evaluate long-term potential and scalability, including the effects of tool upgrades, skill progression, and

unlocked seasonal or rare opportunities. Expanding the experiment to include multiple islands would provide insight into how geographical and resource diversity influences activity performance and overall economic outcomes.

To reduce bias and improve accuracy, a more automated data collection approach could be adopted. This may involve using existing game modifications (mods) or developing custom scripts to track in-game actions, income generation, and resource usage in real time. Automation would minimize human error, standardize measurements across sessions, and allow for more complex analyses of player strategies, seasonal effects, and progression dynamics.

Finally, exploring interactions between multiple activities rather than isolating them individually could reveal synergistic strategies or trade-offs in income optimization, providing a more holistic understanding of the Stardew Valley virtual economy. Incorporating these recommendations would address the current study's limitations, enabling future research to more accurately measure efficiency, strategy effectiveness, and the influence of progression and environmental factors on income outcomes.

VI. Conclusion

This study analyzed income-generating activities in Stardew Valley by empirically comparing Farming, Fishing, Mining, and Foraging under a single-activity, full-year gameplay constraint. The key findings indicate that no single activity universally dominates across all performance metrics, but meaningful distinctions emerge when evaluating daily income and time-based efficiency separately. Fishing yielded the highest mean daily net income (923.72g) and the highest cumulative net income over the observed year, driven by its low capital requirements, consistent daily returns, and limited downside volatility, positioning it as a reliable and stable long-term income strategy. Farming, by contrast, achieved the highest median net income (1,387.50g) and the highest median Gold-Per-Minute (8.46 GPM), reflecting superior time-based efficiency and peak-day productivity on harvest days, though this performance was accompanied by substantial income variability and significant losses during non-harvest periods. Mining demonstrated moderate and episodic performance, with income driven by resource variability and luck sensitivity, while Foraging remained the least profitable and least time-efficient activity across all metrics, suggesting a supportive rather than dominant role in long-term economic optimization. Collectively, these results confirm that economic effectiveness in a constrained gameplay environment depends on player objectives, available capital, and tolerance for income volatility rather than any single universally superior strategy. The observed differences between activities, supported by Shapiro–Wilk normality tests and Kruskal–Wallis non-parametric comparisons with Dunn post-hoc analyses, underscore meaningful trade-offs between cumulative stability and time-based efficiency in gameplay decision-making.

Through conducting this research, the researcher developed a deeper understanding of systematic decision-making and optimization within constrained environments. Isolating variables, tracking longitudinal outcomes, and interpreting distributional trade-offs reinforced the value of disciplined experimental design and data-driven reasoning. More importantly, the study shifted the author's perspective on games from purely entertainment artifacts to structured systems capable of modeling complex economic and strategic behavior in a controlled and analyzable form.

Although the findings are specific to a game environment, they offer conceptual insights applicable beyond the virtual context. The study demonstrates how opportunity cost, specialization, capital investment, and efficiency trade-offs shape outcomes in resource-limited systems. The contrast between Fishing's cumulative income advantage and Farming's superior time efficiency, for instance, mirrors the broader tension between risk-averse, steady-return strategies and high-variance, high-reward approaches found in real-world resource allocation decisions. These principles are relevant to time management, productivity optimization, and strategic planning, where individuals must allocate limited resources to maximize returns under uncertainty.

In conclusion, Stardew Valley serves as a meaningful experimental platform for studying resource management and optimization strategies. By providing longitudinal evidence of activity-specific performance across 448 daily observations and four distinct income-generating activities, this study contributes to the understanding of virtual game economies while illustrating the broader utility of simulated environments as simplified models for economic behavior and decision-making. The findings emphasize that optimal strategies emerge not from singular choices, but from aligning player goals, available resources, and progression stage over time, with trade-offs between income stability, peak efficiency, and cumulative returns carefully considered across different activity types.

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